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(54) **CARGO RESTRAINT SYSTEM**

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USPC 410/46, 71, 77–80, 82, 90–91, 94–95
See application file for complete search history.

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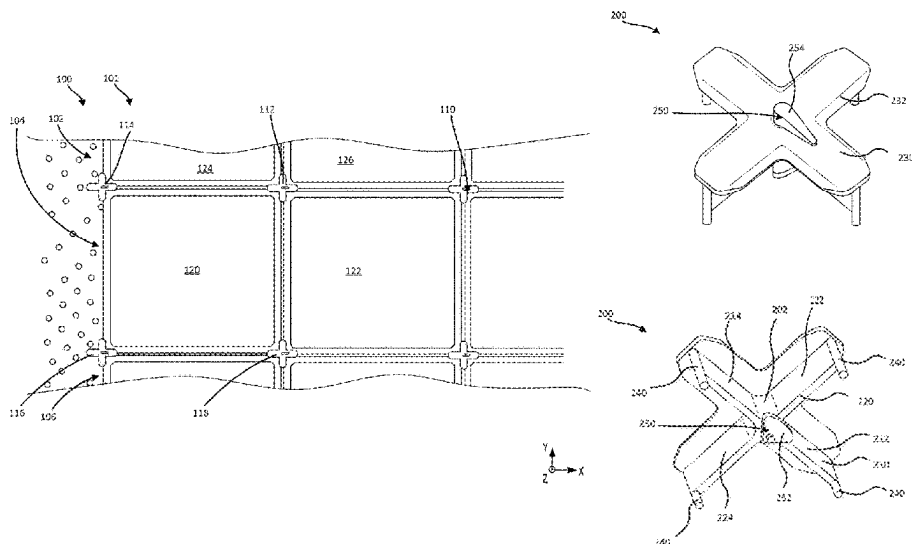
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(57) **ABSTRACT**

A cargo restraint system comprises a plurality of cargo restraint assemblies. Each cargo restraint assembly in the plurality of cargo restraint assemblies may be removably coupled to a cargo deck in various configurations. The plurality of cargo restraint assemblies may be reconfigurable based on a size of a plurality of unit load devices (ULDs) to be loaded and restrained. Each cargo restraint assembly may restrain four or fewer corners of respective ULDs simultaneously.

7 Claims, 9 Drawing Sheets



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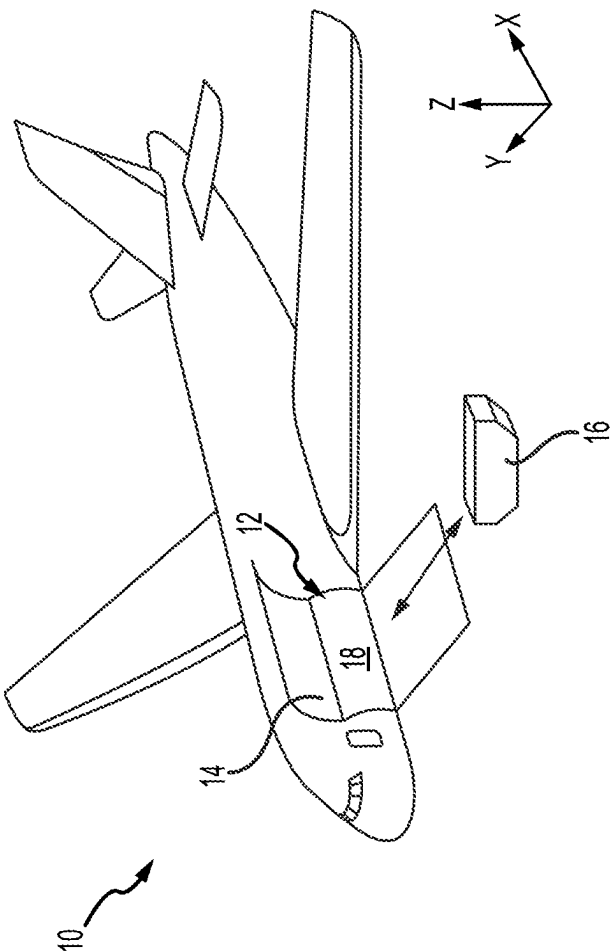
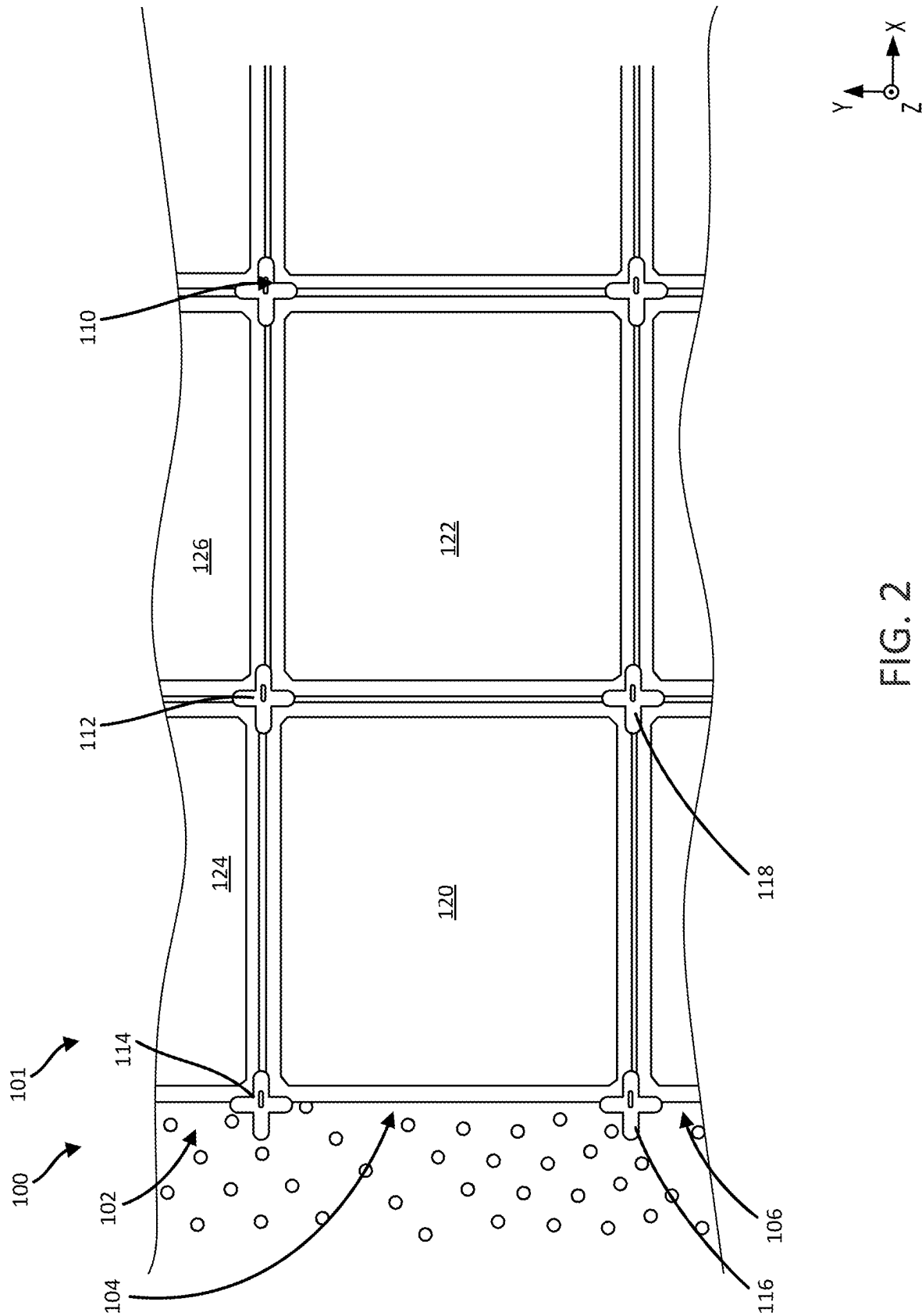


FIG. 1



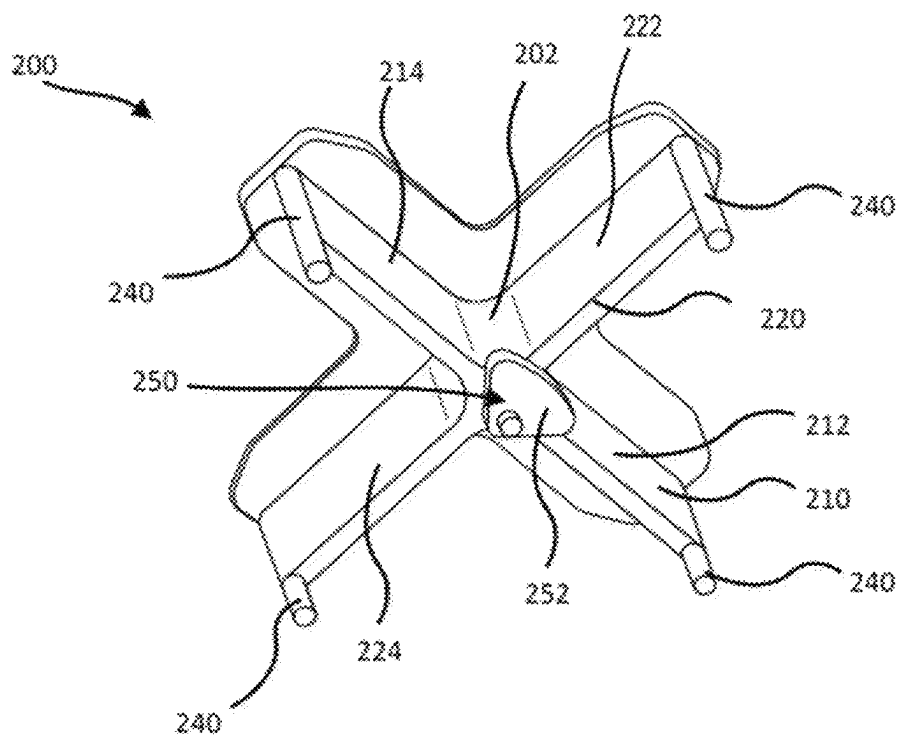
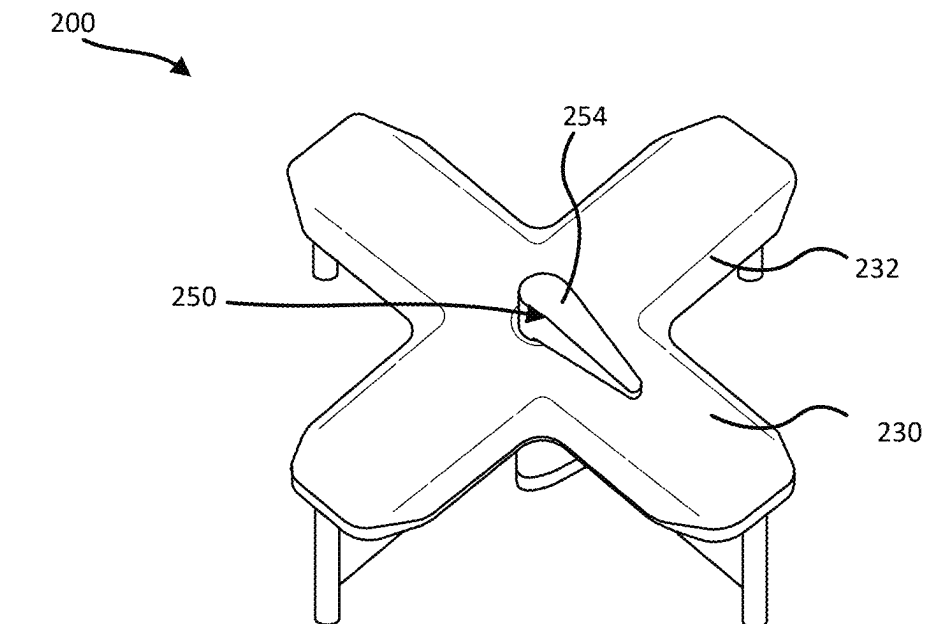


FIG. 3

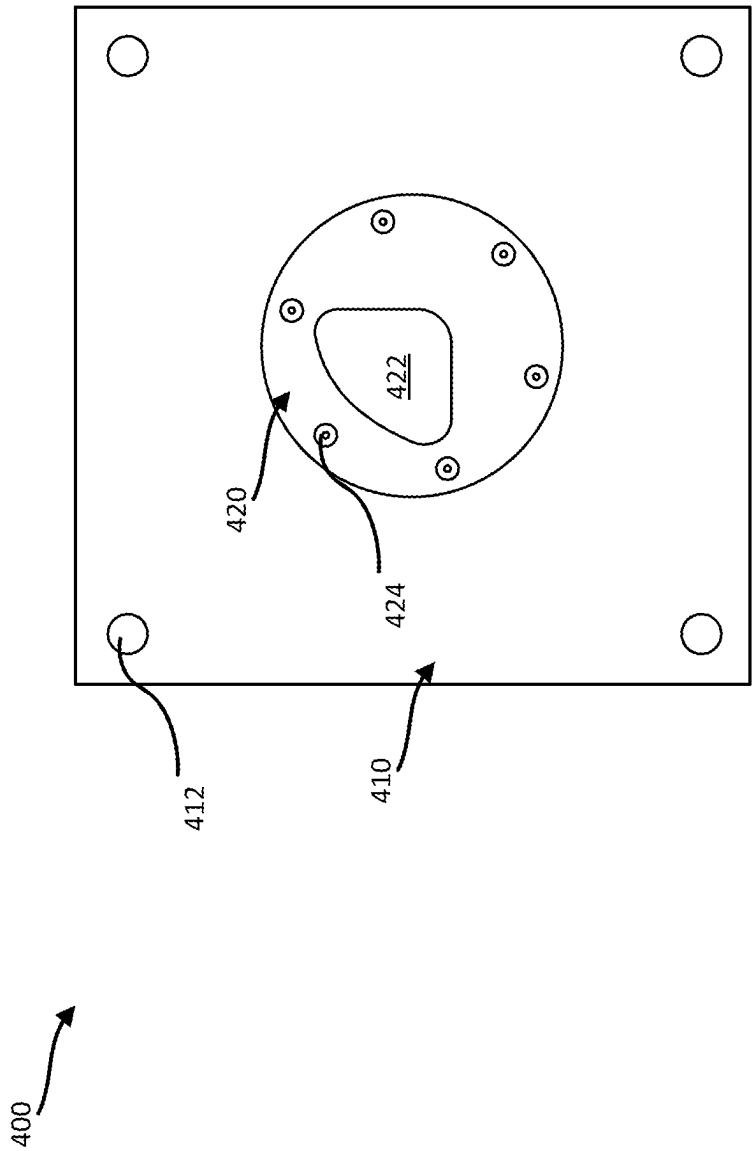


FIG. 4

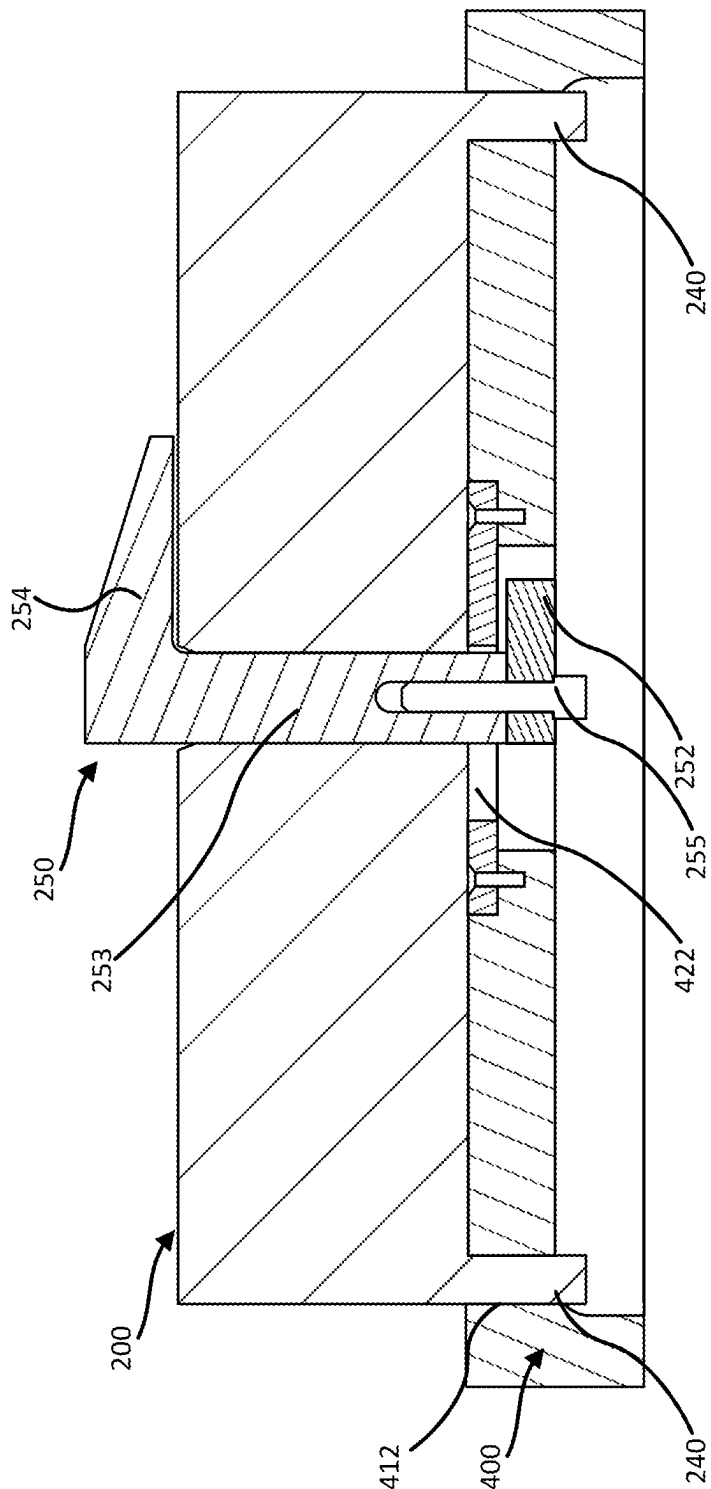


FIG. 5

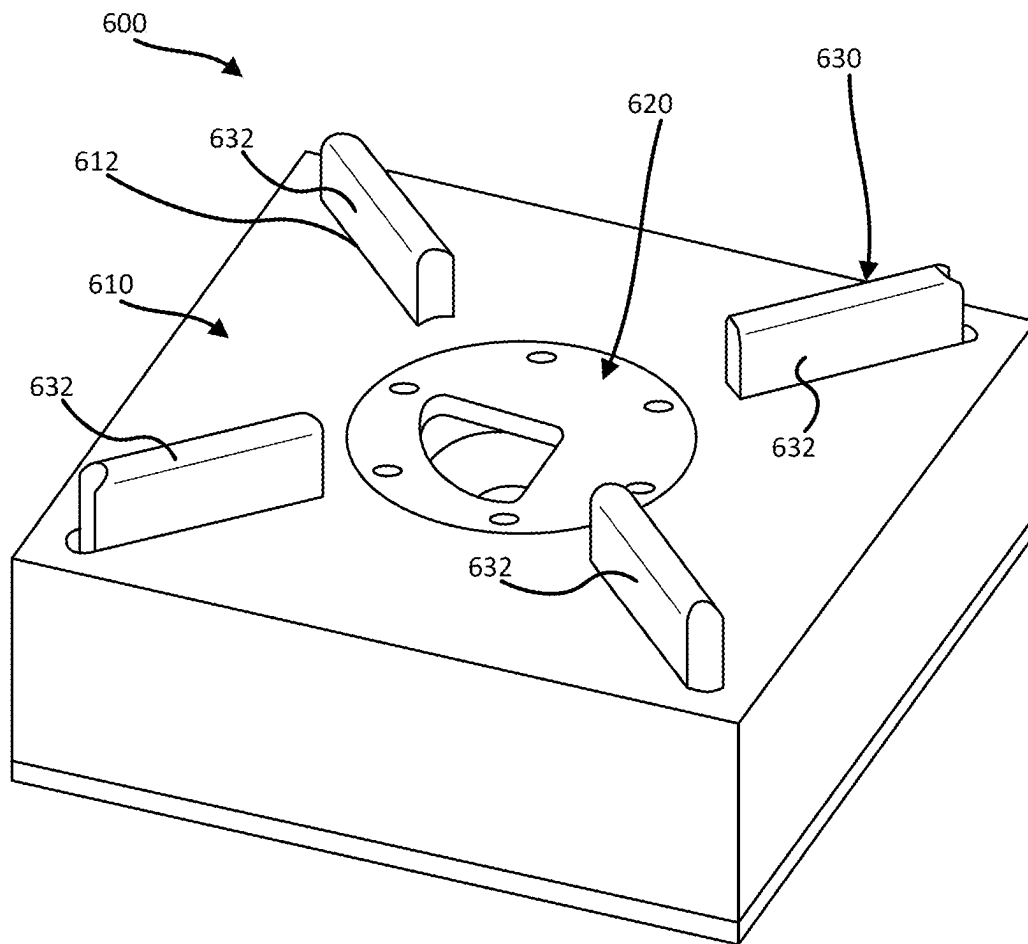
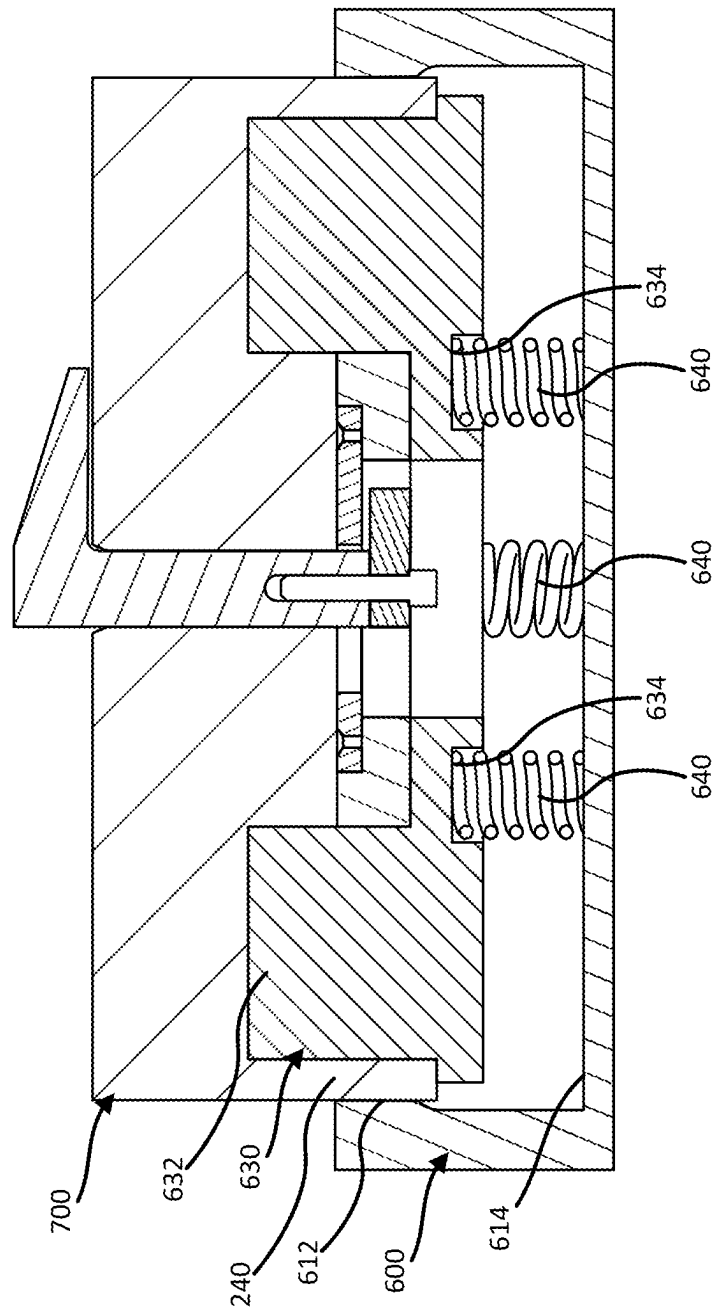


FIG. 6


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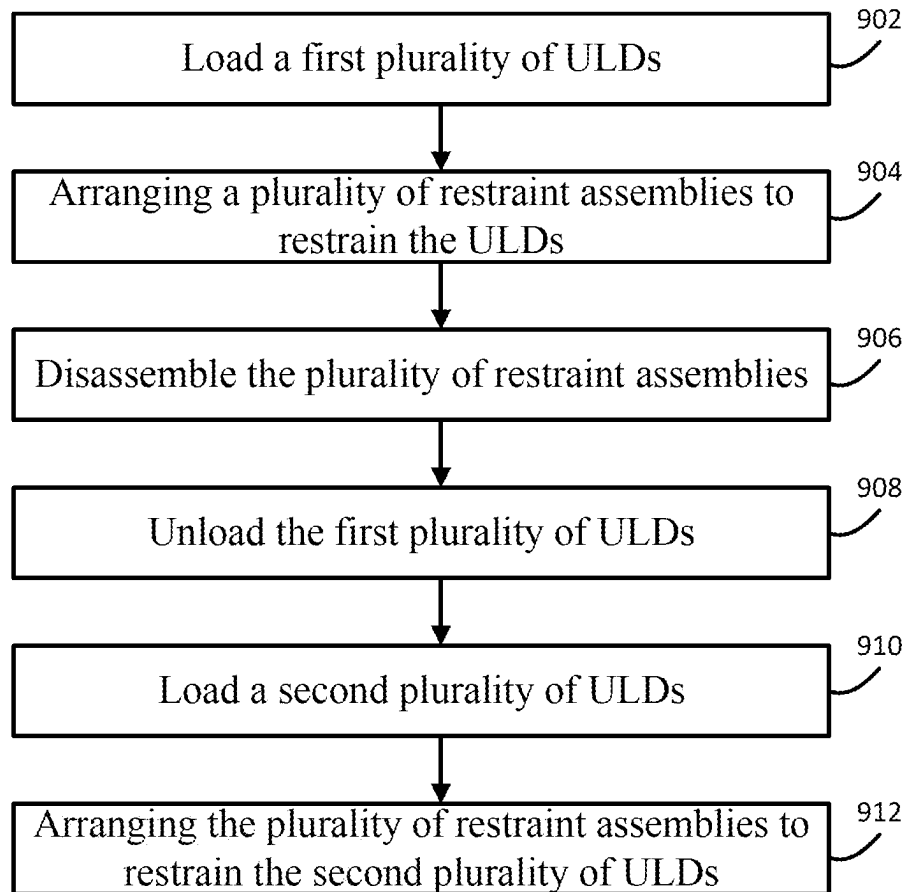


FIG. 9

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CARGO RESTRAINT SYSTEM**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a divisional of, claims priority to and the benefit of, U.S. application Ser. No. 17/179,775 filed Feb. 19, 2021, entitled "CARGO RESTRAINT SYSTEM" (hereinafter the '775 Application). The '775 Application claims priority to and the benefit of U.S. Provisional Application No. 63/025,007, filed May 14, 2020, entitled "CARGO RESTRAINT SYSTEM," (hereinafter the '007 Application). The '775 Application and the '007 Application, are hereby incorporated by reference in their entirety for all purposes.

FIELD

The present disclosure relates generally to cargo handling systems, and more specifically, to a restraint scheme for aircraft cargo handling systems.

BACKGROUND

Aircraft cargo handling systems transport, guide and restrain cargo within the cargo hold of an aircraft. Typical aircraft cargo handling systems use a combination of fixed or moveable guides or restraints to hold cargo pallets or containers (e.g., unit load devices, or ULDs) in place during flight. Most systems use fixed side guides along the outboard edges of the aircraft and/or retractable guides inboard to provide restraint in the Y (lateral) and Z (vertical) directions and retractable restraints that handle the X (longitudinal) and Z directions on the forward and aft edges of the ULDs. Other restraint schemes for other ULDs use fixed side guides with retractable restraints that extend into pockets on the sides of the ULDs to provide X, Y and Z restraint on one or both sides of the ULD and retractable guides along the other side to handle Y and Z restraint.

These typical restraint schemes are not as flexible as they could be in terms of ULD load configuration. These systems can usually accommodate cargo of different sizes in the longitudinal direction by either moving restraints (if the system allows for that) or erecting a different set of restraints, but they are usually not as flexible in the lateral direction because they have lanes of predefined width. Additional hardware (usually fixed to the aircraft) is necessary to accept cargo of a different width. This increases the weight and complexity of the system.

SUMMARY

A cargo restraint assembly is disclosed herein. The cargo restraint assembly may comprise: a center portion; a first longitudinal restraint extending outward from the center portion, the first longitudinal restraint extending vertically from a first proximal end to a first distal end; a first lateral restraint extending outward from the center portion, the first lateral restraint substantially perpendicular to the first longitudinal restraint, the first lateral restraint extending vertically from a second proximal end to a second distal end; a vertical restraint disposed at the first distal end of the first longitudinal restraint, the vertical restraint comprising a flange; a first shear pin disposed at a first outward end of the first longitudinal restraint, the first shear pin extending from the first proximal end away from the first distal end; and a second shear pin disposed at a second outward end of the

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first lateral restraint, the second shear pin extending from the second proximal end away from the second distal end.

In various embodiments, the cargo restraint assembly may further comprise a locking mechanism disposed through the center portion, the locking mechanism configured to couple the cargo restraint assembly to a fitting of a cargo handling system. The cargo restraint assembly may further comprise a second longitudinal restraint and a second lateral restraint. The second longitudinal restraint may extend outward from the center portion in a first opposite direction relative to the first longitudinal restraint, and the second lateral restraint may extend outward from the center portion in a second opposite direction relative to the second longitudinal restraint. The first longitudinal restraint, the second longitudinal restraint, the first lateral restraint, and the second lateral restraint may each comprise a recess disposed therein. The first longitudinal restraint, the second longitudinal restraint, the first lateral restraint, and the second lateral restraint may each comprise a sidewall. The sidewall of the first longitudinal restraint, the second longitudinal restraint, the first lateral restraint, and the second lateral restraint may form an "X" shape. The cargo restraint assembly may further comprise a tab rotating coupled to the center portion, the tab configured to lock the cargo restraint assembly to a fitting of a cargo handling system.

A cargo restraint system for use in a cargo handling system is disclosed herein. The cargo restraint system may comprise: a cargo deck; a plurality of fitting assemblies spaced about the cargo deck; a plurality of cargo restraint assemblies configured to removably couple to the plurality of fitting assemblies, each cargo restraint assembly in the plurality of cargo restraint assemblies configured to restrain at a corner of a unit load device (ULD).

In various embodiments, each cargo restraint assembly in the plurality of cargo restraint assemblies is configured to restrain four or fewer ULDs simultaneously. The plurality of fitting assemblies may be spaced to accommodate ULDs of various sizes. Each cargo restraint assembly in the plurality of cargo restraint assemblies may comprise a first longitudinal restraint, a first lateral restraint, a second longitudinal restraint, a second lateral restraint, and a vertical restraint, each restraint extending outward from a center portion. Each cargo restraint assembly in the plurality of cargo restraint assemblies may comprise a plurality of shear pins disposed at ends distal to the on the first longitudinal restraint, the first lateral restraint, the second longitudinal restraint, and the second lateral restraint, the plurality of shear pins configured to restrict horizontal motion of each cargo restraint assembly during use. Each fitting assembly in the plurality of fitting assemblies may comprise a plurality of apertures configured to receive the plurality of shear pins of a respective cargo restraint assembly in the plurality of cargo restraint assemblies. Each cargo restraint assembly in the plurality of cargo restraint assemblies may comprise a locking mechanism disposed through the center portion, the locking mechanism configured to couple to a respective fitting assembly in the plurality of fitting assemblies. Each fitting assembly in the plurality of fitting assemblies may comprise a spring loaded guide configured to press each cargo restraint assembly against a respective fitting assembly and keep each cargo restraint assembly tight against the respective fitting assembly.

A method for using a cargo restraint system is disclosed herein. The method may comprise: arranging a plurality of restraint assemblies in a first configuration about a cargo deck to restrain a first plurality of Unit Load Devices (ULDs); decoupling the plurality of restraint assemblies

from the cargo deck; and re-arranging the plurality of restraint assemblies in a second configuration about the cargo deck to restrain a second plurality of ULDs, the second configuration being different than the first configuration.

In various embodiments, the first plurality of ULDs are different sizes than the second plurality of ULDs. Each cargo restraint assembly in the plurality of cargo restraint assemblies may be configured to restrain a respective corner of four or fewer ULDs in the first plurality of ULDs simultaneously. Arranging the plurality of restraint assemblies in the first configuration may further comprise arranging the first plurality of ULDs in a grid pattern. Arranging the plurality of restraint assemblies in the first configuration may further comprise disposing the plurality of restraint assemblies at respective corners of the first plurality of ULDs.

The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the detailed description and claims when considered in connection with the drawing figures.

FIG. 1 illustrates an axonometric view of an aircraft being loaded with cargo, in accordance with various embodiments;

FIG. 2 illustrates a top down view of a ULD located on a cargo handling system, in accordance with various embodiments;

FIG. 3 illustrates perspective views of a cargo restraint assembly for a cargo restraint system, in accordance with various embodiments;

FIG. 4 illustrates a top down view of a fitting assembly for a cargo restraint system, in accordance with various embodiments;

FIG. 5 illustrates a cross-sectional view of a cargo restraint assembly coupled to a fitting assembly, in accordance with various embodiments;

FIG. 6 illustrates a perspective view of a fitting assembly for a cargo restraint system, in accordance with various embodiments

FIG. 7 illustrates a perspective view of a cargo restraint assembly coupled to a fitting assembly, in accordance with various embodiments;

FIG. 8 illustrates a cross-sectional view of a cargo restraint assembly coupled to a fitting assembly, in accordance with various embodiments; and

FIG. 9 illustrates a method of using a cargo restraint system, in accordance with various embodiments.

DETAILED DESCRIPTION OF THE DRAWINGS

The detailed description of various embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical mechanical changes may be made without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for pur-

poses of illustration only and not of limitation. For example, the steps recited in any of the method or process descriptions may be executed in any order and are not necessarily limited to the order presented. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full, and/or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact.

Throughout the present disclosure, like reference numbers denote like elements. Accordingly, elements with like element numbering may be shown in the figures but may not necessarily be repeated herein for the sake of clarity.

As used herein, “aft” refers to the direction associated with the tail (i.e., the back end) of an aircraft, or generally, to the direction of exhaust of a gas turbine engine. As used herein, “forward” refers to the direction associated with the nose (i.e., the front end) of an aircraft, or generally, to the direction of flight or motion.

Cargo restraint systems, as disclosed herein, may comprise cargo restraint assemblies configured to guide and restrain the movement of ULDs or other cargo across the floor or “cargo deck” of an aircraft. In accordance with various embodiments, the cargo restraint assemblies may be coupled to a cargo deck and be configured to restrain a ULD in a longitudinal, lateral, and vertical direction. In this regard, the cargo restraint assembly may restrain a first cargo, a second cargo, a third cargo, and a fourth cargo, in accordance with various embodiments. The cargo restraint assembly may be disposed at a corner of the first cargo, the second cargo, the third cargo and the fourth cargo. The restraint assembly may be easily repositioned in the cargo handling system to accommodate different sizes of ULDs to be restrained in a grid, or the like. The cargo restraint assembly may be removably coupled to fittings disposed in the cargo deck and/or stored when not in use.

With reference to FIG. 1, an aircraft 10 is illustrated. Aircraft 10 includes a cargo compartment 12. A cargo door 14 provides access to cargo compartment 12 from outside aircraft 10. Cargo 16 (e.g., pallets or ULDs) may be loaded and unloaded through cargo door 14 and onto a cargo deck 18 of aircraft 10.

Referring now to FIG. 2, a top down view of a cargo restraint system 101 for a cargo handling system 100, in accordance with various embodiments, is illustrated. Cargo handling system 100 may be used to load, move, and unload cargo 16 from cargo deck 18. Cargo handling system 100 includes components configured to facilitate translation of cargo 16 along cargo deck 18. For example, cargo handling system a plurality of roller trays (caster trays), one or more rollers oriented in a longitudinal direction, one or more power drive units (PDUs) configured to propel cargo 16 in a desired direction, and/or one or more ball transfer units (BTUs) configured to translate cargo in a lateral and longitudinal direction. As used herein, the term “longitudinal” refers to forward and aft directions (i.e., a direction parallel to the X-axis on the provided X-Y-Z axes). As used herein, the term “lateral” refers to directions perpendicular to the longitudinal direction (i.e., directions parallel to the y-axis on the provided X-Y-Z axes). As used herein, the term “vertical” refers to directions perpendicular to cargo deck 18 (i.e., direction perpendicular to the X-Y plane and/or parallel to the Z-axis on the provided X-Y-Z axes).

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In various embodiments, the cargo handling system **100** may include a first cargo row **102**, a second cargo row **104**, and a third cargo row. The first cargo row **102** may be disposed laterally adjacent to the second cargo row **104**, and the third cargo row **106** may be disposed laterally adjacent to the second cargo row **104**. Although illustrated as comprising three cargo rows, any number of cargo rows is within the scope of this disclosure.

In various embodiments, the cargo restraint system **101** may comprise a plurality of cargo restraint assemblies **110**, as described further herein. In various embodiments, the cargo restraint system **101** is configured to restrain each of the four corners of a respective ULD. For example, the cargo restraint system **101** may comprise a first cargo restraint assembly **112** disposed at a first corner of a ULD **120**, a second cargo restraint assembly **114** disposed at a second corner of the ULD **120**, a third cargo restraint assembly **116** disposed at a third corner of the ULD **120**, and a fourth cargo restraint assembly **118** disposed at a fourth corner of the ULD **120**. In various embodiments, each cargo restraint assembly in the plurality of cargo restraint assemblies **110** may be configured to restrain up to four corners of four different ULDs simultaneously. For example, the first cargo restraint assembly **112** may be configured to restrain ULDs **120**, **122** from the second cargo row **104** and ULDs **124**, **126** from the first cargo row **102** simultaneously.

In various embodiments, each ULD comprises a flange extending outward from a main body of the ULD. For example, the first ULD **120** comprises a flange **121** extending outward around a perimeter of a main body **123** of the flange **121**. In this regard, the flange **121** may interface with the restraint assemblies (e.g., restraint assemblies) **112**, **114**, **116**, **118**, in accordance with various embodiments.

Referring now to FIG. 3, perspective views of a cargo restraint assembly **200** is illustrated, in accordance with various embodiments. In various embodiments, the plurality of cargo restraint assemblies **110** from FIG. 2 may be in accordance with the cargo restraint assembly **200**. In various embodiments, the cargo restraint assembly **200** is configured to restrain at least one ULD from FIG. 2. As such, although disclosed herein as restraining four corners simultaneously, a restraint assembly configured to restrain only a single corner of only one ULD is within the scope of this disclosure. In various embodiments, the cargo restraint assembly comprises a first longitudinal restraint **210**, a first lateral restraint **220**, a vertical restraint **230**, at least two shear pins **240**, and a locking mechanism **250**.

In various embodiments, the first longitudinal restraint **210** comprises a sidewall **212** extending vertically from a proximal end to a distal end. Similarly, the first lateral restraint **220** comprises a sidewall **222** extending vertically from a proximal end to a distal end. The sidewall **212** of the first longitudinal restraint **210** may be substantially perpendicular to the sidewall **222** of the first lateral restraint **220**. "Substantially perpendicular," as disclosed herein is perpendicular ± 10 degrees. The sidewalls **212**, **222** may extend outward from a center portion **202**. In various embodiments, the first lateral restraint **220** is configured to restrain a respective ULD in a lateral direction when in use in a cargo restraint system (e.g., cargo restraint system **101** from FIG. 2). Similarly, the first longitudinal restraint **210** is configured to restrain a respective ULD in a longitudinal direction when in use in a cargo restraint system (e.g., cargo restraint system **101** from FIG. 2).

In various embodiments, the vertical restraint **230** is disposed at the distal end of the sidewall **212** of the first longitudinal restraint **210** and the distal end of the sidewall

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222 of the first lateral restraint **220**. In various embodiments, the vertical restraint **230** comprises a flange **232** extending horizontally from the distal end of the sidewalls **212**, **222**. In various embodiments, the flange **232** may form a complimentary shape to a corner of a ULD. In various embodiments, the vertical restraint **230** of the cargo restraint assembly **200** is configured to restrain a ULD in the vertical direction. For example, a ULD may comprise a tab that extends under the vertical restraint **230**, when the cargo restraint system **101** from FIG. 2 is being utilized, and tab may be restrained in the vertical direction by the vertical restraint **230**.

In various embodiments, the locking mechanism **250** may be operably coupled to the center portion **202**. The locking mechanism **250** may be any mechanism configured to lock a cargo restraint assembly **200** to a cargo deck, a fitting for a cargo deck, or the like. Although illustrated herein as comprising a rotatable tab, any locking mechanism is within the scope of this disclosure. In various embodiments, the locking mechanism comprises a tab **252** disposed proximate the proximal ends of sidewalls **212**, **222** and center portion **202**. The tab **252** may be rotatably coupled to a handle **254** disposed proximate the vertical restraint **230**. In this regard, the tab **252** may be configured to be disposed through a complimentary aperture in a fitting or a cargo deck and rotated to lock the restraint assembly **200** vertically in place.

In various embodiments, the cargo restraint assembly **200** may further comprise a second longitudinal restraint **214** and a second lateral restraint **224**. The second longitudinal restraint **214** may extend outward from the center portion **202** in a direction opposite to the first lateral restraint **220**. Similarly, the second lateral restraint **224** may extend outward from the center portion **202** in a direction opposite to the first lateral restraint **220**. In various embodiments, the longitudinal restraints **210**, **214** and the lateral restraints **220**, **224** may form a substantially "X" shape. Each sidewall of the restraints **210**, **214**, **220**, **224** may be substantially equidistant, in accordance with various embodiments. In various embodiments the sidewalls may have different distances and configured for a specific orientation if mistake proofing is desired. In various embodiments, by having the same distance, assembling the cargo restraint assembly **200** in a restraint system (e.g., cargo restraint system **101** from FIG. 2). The second longitudinal restraint **214** may be in accordance with the first longitudinal restraint **210** and the second lateral restraint **224** may be in accordance with the first lateral restraint **220**.

In various embodiments, each sidewall of the restraints **210**, **214**, **220**, **224** includes a shear pins **240** disposed distal to the center portion **202** of the cargo restraint assembly **200**. The shear pin **240** for each sidewall of the restraints **210**, **214**, **220**, **224** extends in a vertical direction away from proximal end in a downward direction away from the respective distal end for the respective sidewall. The shear pins **240** are configured to prevent horizontal translation of the (e.g., translation in the X-Y plane from FIG. 2), while the cargo restraint assembly **200** is in use.

Referring now to FIG. 4, a top down view of a fitting assembly **400**, in accordance with various embodiments, is illustrated. The fitting assembly **400** may be attached to a cargo deck (e.g., cargo deck **18** from FIG. 1), or a separate component configured to be coupled to a cargo deck. In various embodiments, the fitting assembly **400** may comprise a main body **410** and a plate **420**. The plate **420** may be removably coupled to the main body **410**. In this regard, the plate **420** may be easily replaced if the plate **420** wears or becomes unusable after use, in accordance with various

embodiments. In various embodiments, the plate **420** may be integral with the main body **410**. In various embodiments, the main body comprises apertures **412** disposed in complimentary manner to shear pins **240** from the cargo restraint assembly **200**. In this regard, apertures **412** are configured to receive shear pins **240** therein and prevent translation in a horizontal direction of a respective restraint assembly (e.g., cargo restraint assembly **200** from FIG. 3).

In various embodiments, the plate **420** comprises a tab aperture **422** complimentary in shape to the tab **252** of the locking mechanism **250** from FIG. 3. In this regard, the tab aperture **422** may be configured to receive a respective tab therethrough, and the tab may be rotated to lock the respective restraint assembly (e.g., cargo restraint assembly **200** from FIG. 3) in place. The plate **420** may be coupled to the main body by any method known in the art (e.g., fasteners **424** or the like).

Referring now to FIG. 5, a cross sectional view of a cargo restraint assembly **200** coupled to a fitting assembly **400** is illustrated, in accordance with various embodiments. In various embodiments, the shear pins **240** are disposed within apertures **412** of the fitting assembly **400**. Similarly, the tab **252** of the locking mechanism **250** is disposed through the tab aperture **422** and rotated to an opposite orientation of the tab aperture **422**. In this regard, the tab **252** may prevent vertical translation of the cargo restraint assembly **200** when the cargo restraint assembly **200** is in use in a cargo restraint system (e.g., cargo restraint system **101** from FIG. 2).

In various embodiments, the locking mechanism **250** may further comprise a shaft extending vertically from the handle **254** through the center portion of the cargo restraint assembly **200** to the tab **252**. In various embodiments, the tab **252** may be coupled to the shaft **253** by any method known in the art (e.g., a fastener **255**, a weld, a braze, a press fit, or the like).

In various embodiments, the shear pins **240** may also transmit longitudinal and lateral shear loads through the fitting assembly **400** and into the aircraft structure. In various embodiments, the shear pins **240** may be sized and configured to engage apertures **412** prior to the tab **252** engaging the tab aperture **422**. In this regard, proper alignment may be ensured and/or the restraint assembly may stay in place while the handle **254** is rotated to lock the cargo restraint assembly **200** in place.

Referring now to FIG. 6, a perspective view of a fitting assembly **600**, in accordance with various embodiments, is illustrated. The fitting assembly **600** may include a main body **610** and a plate **620**. The main body **610** may be in accordance with main body **610** with the addition of slots **612** extending from a radially inner end proximate the plate **620** radially outward. In various embodiments, the plate **620** may be in accordance with the plate **420** from FIG. 4.

In various embodiments, the fitting assembly **600** may further comprise a spring loaded guide **630**. The spring loaded guide **630** may comprise protrusions **632** configured to extend through the slots **612**. The protrusions **632** may be configured to receive respective sidewall from a lateral or longitudinal restraint of a respective restraint assembly. The respective sidewall may comprise a recess complementary in shape to the protrusions **632**.

Referring now to FIG. 7, a perspective view of a restraint assembly **700** coupled to a fitting assembly **600**, in accordance with various embodiments. The restraint assembly **700** may be in accordance with cargo restraint assembly **200** and further comprise recesses **710** disposed in restraints **210**, **214**, **220**, **224** from cargo restraint assembly **200** in FIG. 2. The recesses **710** may be complimentary in shape to the

protrusions **632** of the spring loaded guide **630** of fitting assembly **600**. In various embodiments, the spring loaded guide **630** is configured to hold a respective ULD in place while an operator is installing the restraint. In various embodiments, the spring loaded guide **630** may be pushed below the floor when a respective ULD is being loaded over the fitting assembly **600** or when the ULD is being restrained over the fitting assembly **600**.

Referring now to FIG. 8, a cross-sectional view of the restraint assembly **700** coupled to the fitting assembly **600** is illustrated, in accordance with various embodiments. In various embodiments, the fitting assembly **600** may further comprise springs **640** disposed between a surface **614** of main body **610** and recesses **634** disposed in a body of the spring loaded guide **630**. In various embodiments, the shear pins **240** of the restraint assembly **700** may be disposed between slots **612** of fitting assembly **600** and protrusions **632** of spring loaded guide **630**. In this regard shear pins **240** may still prevent horizontal translation of the restraint assembly **700** during use of the restraint assembly **700** in a cargo restraint system (e.g., cargo restraint system **101** from FIG. 1).

Referring now to FIG. 9, a method of using a cargo restraint system (e.g., cargo restraint system **101** from FIG. 1), is illustrated, in accordance with various embodiments. The method **900** comprises loading a first plurality of ULDs (step **902**). The first plurality of ULDs may be loaded via a cargo handling system (e.g., cargo handling system **100** from FIG. 1). The first plurality of ULDs may be a first size.

The method **900** further comprises arranging a plurality of restraint assemblies to restrain the ULDs (step **904**). The restraint assemblies may be assembled in accordance with the cargo restraint system **101** from FIG. 1 (e.g., arranged at corners of the ULDs and restraining between 1 and 4 corners simultaneously).

The method **900** further comprises disassembling the plurality of restraint assemblies (step **906**) and unloading the first plurality of ULDs (step **908**). The method further comprises loading a second plurality of ULDs (step **910**). The second plurality of ULDs may comprise a second size. The second size may be a different size than the first size of the first plurality of ULDs. The method further comprises arranging the plurality of restraint assemblies to restrain the second plurality of ULDs (step **912**). The restraint assemblies may be placed in different locations compared to the first plurality of ULDs.

In various embodiments, a restraint system as disclosed herein (e.g., cargo restraint system **101**) may allow more flexibility in a ULD load configuration of a cargo handling system relative to typical cargo handling systems. The cargo deck may have fitting assemblies (e.g., fitting assembly **400** or fitting assembly **600**) at regular intervals and configurable for ULDs of various sizes. In various embodiments, the cargo restraint assembly may save on system weight, complexity, and/or part count relative to typical cargo restraint assemblies. In this regard, in accordance with various embodiments, the restraints may be repositioned and used elsewhere in the compartment rather than having multiple sets of restraint assemblies fixed to the floor throughout the compartment to accommodate all different ULD sizes.

Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or

physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” It is to be understood that unless specifically stated otherwise, references to “a,” “an,” and/or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. All ranges and ratio limits disclosed herein may be combined.

Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Elements and steps in the figures are illustrated for simplicity and clarity and have not necessarily been rendered according to any particular sequence. For example, steps that may be performed concurrently or in different order are illustrated in the figures to help to improve understanding of embodiments of the present disclosure.

Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element is intended to invoke 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus

that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

What is claimed is:

1. A cargo restraint assembly, comprising:

a center portion;

a first longitudinal restraint extending outward from the center portion, the first longitudinal restraint extending vertically from a first proximal end to a first distal end;

a first lateral restraint extending outward from the center portion, the first lateral restraint substantially perpendicular to the first longitudinal restraint, the first lateral restraint extending vertically from a second proximal end to a second distal end;

a vertical restraint disposed at the first distal end of the first longitudinal restraint, the vertical restraint comprising a flange;

a first shear pin disposed at a first outward end of the first longitudinal restraint, the first shear pin extending from the first proximal end away from the first distal end; and

a second shear pin disposed at a second outward end of the first lateral restraint, the second shear pin extending from the second proximal end away from the second distal end.

2. The cargo restraint assembly of claim 1, further comprising a locking mechanism disposed through the center portion, the locking mechanism configured to couple the cargo restraint assembly to a fitting of a cargo handling system.

3. The cargo restraint assembly of claim 1, further comprising a second longitudinal restraint and a second lateral restraint.

4. The cargo restraint assembly of claim 3, wherein the second longitudinal restraint extends outward from the center portion in a first opposite direction relative to the first longitudinal restraint, and wherein the second lateral restraint extends outward from the center portion in a second opposite direction relative to the first lateral restraint.

5. The cargo restraint assembly of claim 4, wherein the first longitudinal restraint, the second longitudinal restraint, the first lateral restraint, and the second lateral restraint each comprise a recess disposed therein.

6. The cargo restraint assembly of claim 3, wherein the first longitudinal restraint, the second longitudinal restraint, the first lateral restraint, and the second lateral restraint each comprise a sidewall, wherein the sidewall of the first longitudinal restraint, the second longitudinal restraint, the first lateral restraint, and the second lateral restraint form an “X” shape.

7. The cargo restraint assembly of claim 1, further comprising a tab rotatably coupled to the center portion, the tab configured to lock the cargo restraint assembly to a fitting of a cargo handling system.

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